



Digital immortality

Researchers are working on digital resurrections of departed loved ones based on algorithms derived from their emails and messages. <https://digit.in/bringback>



AI predicts life on other worlds

The probability of life on other worlds can be calculated by AI, the study uses artificial neural networks (ANNs). <https://digit.in/aipredict>

ELECTRONS TRAVELLING FASTER THAN LIGHT IN GLASS

Don't worry, no fundamental laws of physics were broken, and Einstein's theory of relativity remains as sound as ever.

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hen Einstein came up with his special theory of relativity, he proposed what is perhaps one of the

most famous equations known to man. According to $E=mc^2$, which is a cornerstone of how scientists currently understand the universe, no particles with mass can travel faster than the light in vacuum. So far, every known phenomenon in the universe has toed the line when it comes to this rule. However, inside a material like

glass or water, particles such as electrons can travel faster than light within the same medium. In such a situation, a special type of radiation is emitted by the medium known as Cherenkov radiation. This radiation has now allowed researchers from the Tata Institute of Fundamental Research (TIFR), the Rutherford Appleton Laboratory in UK and

Centre for Intense Laser Studies and Applications (CELIA) in France to track the life history of such 'faster than light' electrons.

Pavel Cherenkov, a Soviet Scientist was the first to experimentally observe the radiation in a lab environment. Under the direction of Sergey Vavilov, Cherenkov studied the radiation by irradiating various liquids with gamma rays from a radium source. The phenomenon is also known as the Vavilov-Cherenkov radiation for this reason. It was not the first time that researchers had come across the bluish glow though. There is evidence to suggest that Pierre and Marie Curie were familiar with the glow from glass containers with radium salts within them. Il'ja Frank and Igor Tamm theoretically studied the phenomenon, and developed an explanation that was consistent with Einstein's Special Theory of Relativity. For their work, Cherenkov, Frank and Tamm shared the 1958 Nobel Prize in physics "for the discovery and the interpretation of the Cherenkov effect".



The Cherenkov Effect in the Reed Research Reactor